



SRKR ENGINEERING COLLEGE(A)::BHIMAVARAM

Department of Computer Science and Engineering

IV/IV B.Tech Ist Semester

Assignment-1

ESTD:1980 Academic Year-2026-2027	
Program: B. Tech (AI & ML)	Section: A, B, C
Name of the Course: GENERATIVE AI	Course Code: B23AM4101
Max Marks: 5M	

Answer All the Questions

S. No.	Question	CO	KL	M
1.	A company wants to develop three AI applications: generating product descriptions, creating realistic product images, and reconstructing noisy images using a latent representation. Apply your knowledge of Autoregressive Models, GANs, and VAEs to select an appropriate model for each application and justify your choices.	1	K3	1
2.	An autoregressive model assigns the probabilities $P(AI)=0.8$, $P(\text{generates} AI)=0.6$, and $P(\text{text} AI, \text{generates})=0.7$. Calculate the probability of generating the sequence "AI generates text."	1	K3	1
3.	Apply the RAG process and show the main steps the system should follow before generating the answer for the user query, "What are the requirements for a student to be promoted to the next academic year?"	2	K3	1
4.	An LLM answers, "The Eiffel Tower is located in Rome." Find the problem and apply two suitable techniques to reduce the chance of such an incorrect response.	2	K3	1
5.	A student provides an image of a bar chart to a multimodal LLM and asks, "Which category has the highest value?" Demonstrate how the multimodal LLM would use the image and text input to produce the answer.	2	K3	1